Name: Gokul Shiva Kumar (20097813)

Course Title: M.Sc. Cyber Security

Lecturer Name: Swati Dongre

Module/Subject Title: Advanced Programming Techniques

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1. Introduction

This report presents the development and implementation of four programming tasks using C# and Python. The assignment focuses on object-oriented programming concepts, data structures, client-server communication, and web scraping. The project is structured into two main sections: C# applications (Tasks 1 and 2) and Python applications (Tasks 3 and 4). Each task was designed to simulate real-world scenarios and demonstrate practical programming skills.

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2. Task 1: Churros Ordering System (C#)

This task involves the development of a console-based application for a food truck that sells churros. The system allows the stall owner to manage customer orders efficiently using a menu-driven interface.

The application was implemented using object-oriented principles such as classes, objects, encapsulation, constructors, and properties. Two main classes were created: **Order** and **Churros**. The Order class manages attributes such as order details, quantity, bill amount, and order number. Methods such as place_order(), pay_bill(), and collect_order() were implemented to handle core operations. The Churros class defines the available menu items and their prices.

A queue data structure was used to manage customer orders, ensuring a first-in-first-out (FIFO) system. This accurately reflects real-world queue handling in food services. The pay_bill() method calculates the total cost based on selected items and quantity, and a unit test was implemented to validate its correctness.

2.1 Screenshot Placement:

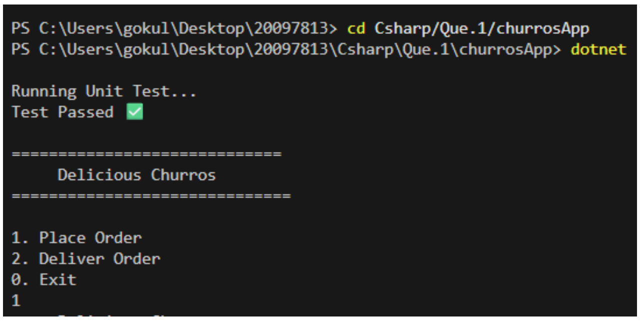

fig 2.1.1: Menu interface

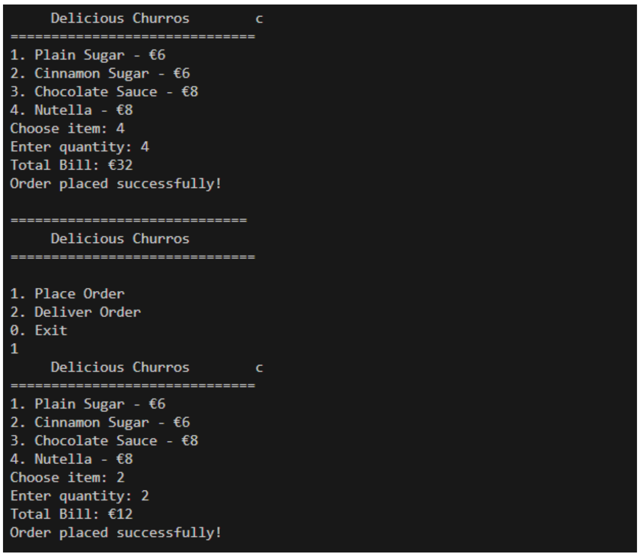

fig 2.1.2: Order placement

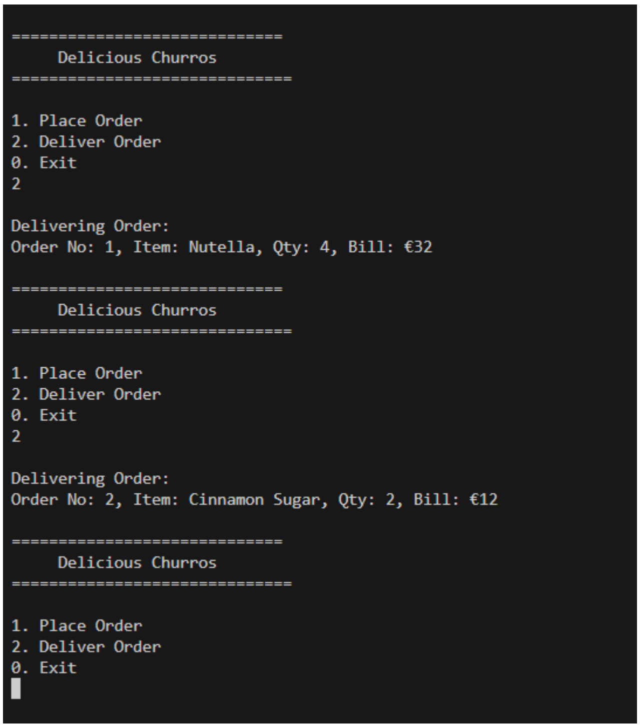

fig 2.1.3: Order delivery process

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3. Task 2: Periodic Table Application (C#)

This task required storing and retrieving information about the first 30 elements of the periodic table. A suitable data structure, such as a dictionary, was used to map atomic numbers to element details.

The program prompts the user to enter an atomic number and displays relevant information, including element name and classification. The application continues running until the user chooses to exit.

The design ensures efficient lookup and retrieval of data, demonstrating the use of data structures in solving real-world problems. Input validation was also implemented to handle incorrect entries.

3.1 Screenshot Placement:

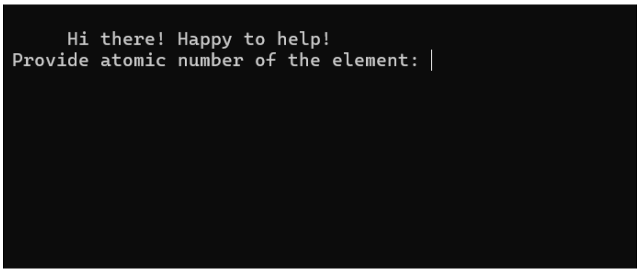


fig 3.1.1: User input (atomic number)

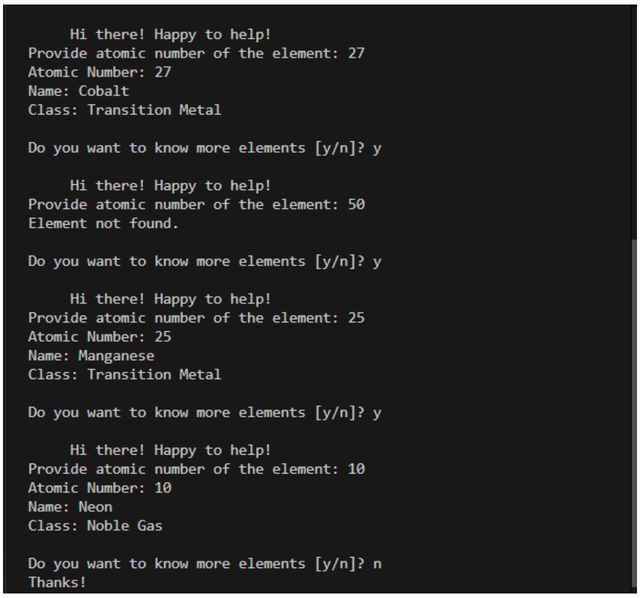


fig 3.1.2: Output displaying element details

4. Task 3: Client-Server Registration System (Python)

This task involves developing a client-server application using TCP protocol. The system simulates a car rental registration process where users provide personal details such as name, address, PPS number, and driving license information.

The client application collects user input and sends it to the server. The server processes the data, stores it in a database (file-based storage), and generates a unique registration ID. This ID is then sent back to the client.

Socket programming was used to establish communication between the client and server. The server runs continuously and handles incoming connections, while the client interacts with the user.

4.1 Screenshot Placement:

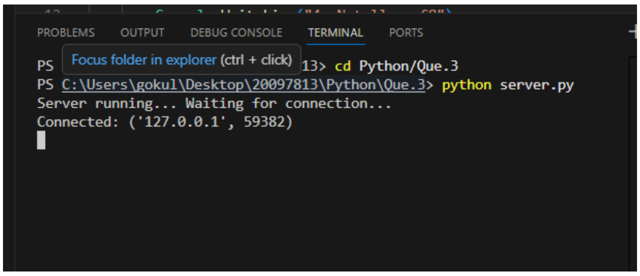


Fig 4.1.1: Server terminal showing connection established

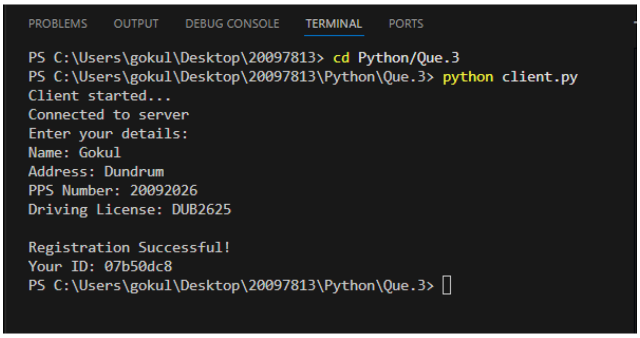
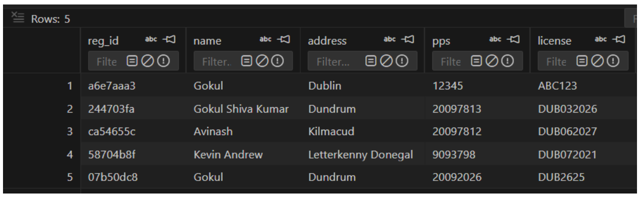


Fig 4.1.2: Client terminal showing user input and received registration ID



Rows: 5

	reg_id	name	address	pps	license
1	a6e7aaa3	Gokul	Dublin	12345	ABC123
2	244703fa	Gokul Shiva Kumar	Dundrum	20097813	DUB032026
3	ca54655c	Avinash	Kilmacud	20097812	DUB062027
4	58704b8f	Kevin Andrew	Letterkenny Donegal	9093798	DUB072021
5	07b50dc8	Gokul	Dundrum	20092026	DUB2625

Fig 4.1.3: Customer Database showing user Data

5. Task 4: Web Scraping Application (Python)

This task focuses on extracting data from a website using Python. The target webpage contains book details such as name, price, and rating. The requests and Beautiful Soup libraries were used to fetch and parse the HTML content.

The program extracts relevant data and stores it in a CSV file named books.csv. After storing the data, the program reads the CSV file and displays the contents in the terminal.

This task demonstrates data extraction, file handling, and automation using Python.

5.1 Screenshot Placement:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\gokul\Desktop\20097813> cd Python/Que.4
PS C:\Users\gokul\Desktop\20097813\Python\Que.4> python scraper.py
Starting scraping...
Books found: 11
Book: It's Only the Himalayas | Price: £45.17 | Rating: Two
Book: Full Moon over Noah's Ark: An Odyssey to Mount Ararat and Beyond
| Price: £49.43 | Rating: Four
Book: See America: A Celebration of Our National Parks & Treasured Site
s | Price: £48.87 | Rating: Three
Book: Vagabonding: An Uncommon Guide to the Art of Long-Term World Trav
el | Price: £36.94 | Rating: Two
Book: Under the Tuscan Sun | Price: £37.33 | Rating: Three
Book: A Summer In Europe | Price: £44.34 | Rating: Two
Book: The Great Railway Bazaar | Price: £30.54 | Rating: One
Book: A Year in Provence (Provence #1) | Price: £56.88 | Rating: Four
Book: The Road to Little Dribbling: Adventures of an American in Britai
n (Notes From a Small Island #2) | Price: £23.21 | Rating: One
Book: Neither Here nor There: Travels in Europe | Price: £38.95 | Ratin
g: Three
Book: 1,000 Places to See Before You Die | Price: £26.08 | Rating: Five

Data saved to books.csv

Reading from CSV:

['Book Name', 'Price', 'Rating']
["It's Only the Himalayas", '£45.17', 'Two']
['Full Moon over Noah's Ark: An Odyssey to Mount Ararat and Beyond', '£
49.43', 'Four']
['See America: A Celebration of Our National Parks & Treasured Sites',
'£48.87', 'Three']
['Vagabonding: An Uncommon Guide to the Art of Long-Term World Travel',
 '£36.94', 'Two']
['Under the Tuscan Sun', '£37.33', 'Three']
['A Summer In Europe', '£44.34', 'Two']
['The Great Railway Bazaar', '£30.54', 'One']
['A Year in Provence (Provence #1)', '£56.88', 'Four']
['The Road to Little Dribbling: Adventures of an American in Britain (N
otes From a Small Island #2)', '£23.21', 'One']
['Neither Here nor There: Travels in Europe', '£38.95', 'Three']
['1,000 Places to See Before You Die', '£26.08', 'Five']
PS C:\Users\gokul\Desktop\20097813\Python\Que.4>
```

Fig 5.1.1: Terminal output showing scraped data

A10 | The Road to Little Dribbling: Adventures of ar

	A	B	C
1	Book Name	Price	Rating
2	It's Only the Himalayas	Â£45.17	Two
3	Full Moon over Noahâ€™s Ark: An Odyssey to Mount Ararat and Beyond	Â£49.43	Four
4	See America: A Celebration of Our National Parks & Treasured Sites	Â£48.87	Three
5	Vagabonding: An Uncommon Guide to the Art of Long-Term World Travel	Â£36.94	Two
6	Under the Tuscan Sun	Â£37.33	Three
7	A Summer In Europe	Â£44.34	Two
8	The Great Railway Bazaar	Â£30.54	One
9	A Year in Provence (Provence #1)	Â£56.88	Four
10	The Road to Little Dribbling: Adventures of an American in Britain (Notes From a Small Island #2)	Â£23.21	One
11	Neither Here nor There: Travels in Europe	Â£38.95	Three
12	1,000 Places to See Before You Die	Â£26.08	Five
13			

Fig 5.1.2: CSV file content (books.csv)

6. GitHub and CI/CD Integration

The project was managed using Git and hosted on GitHub. Multiple commits were made to track progress throughout development. A GitHub Actions workflow was implemented to automate the build and execution process.

The workflow includes:

- Running the Python scraper
- Building the C# application
- Ensuring code integrity on every push

This demonstrates knowledge of version control and continuous integration practices.

6.1 Screenshot Placement:

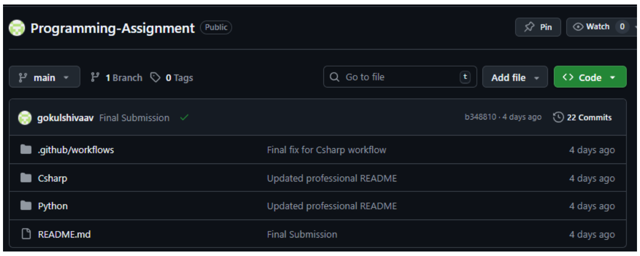


fig 6.1.1: GitHub repository

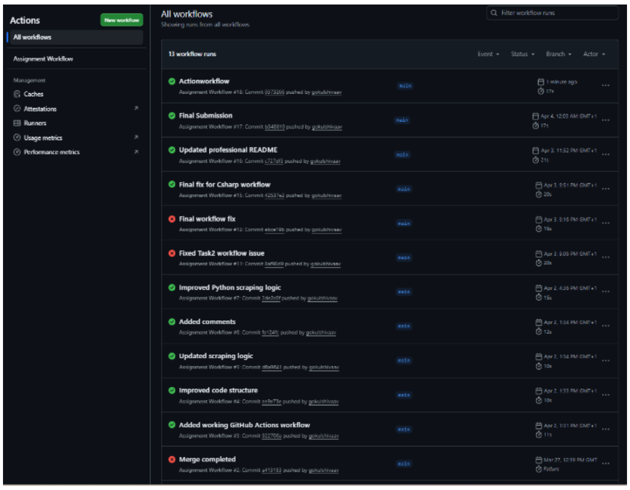


Fig 6.1.2: Actions tab showing successful workflow (green status)

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7. Conclusion

This assignment successfully demonstrates the application of programming concepts across multiple domains. C# was used to implement structured, object-oriented applications, while Python was used for networking and data extraction tasks. The integration of GitHub Actions further enhances the project by introducing automation and continuous integration.

Overall, the project highlights problem-solving skills, software design principles, and the ability to work with multiple technologies effectively.

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8. References:

9. GitHub Link:

https://github.com/gokulshivaav/Programming-Assignment.git